Problem 1

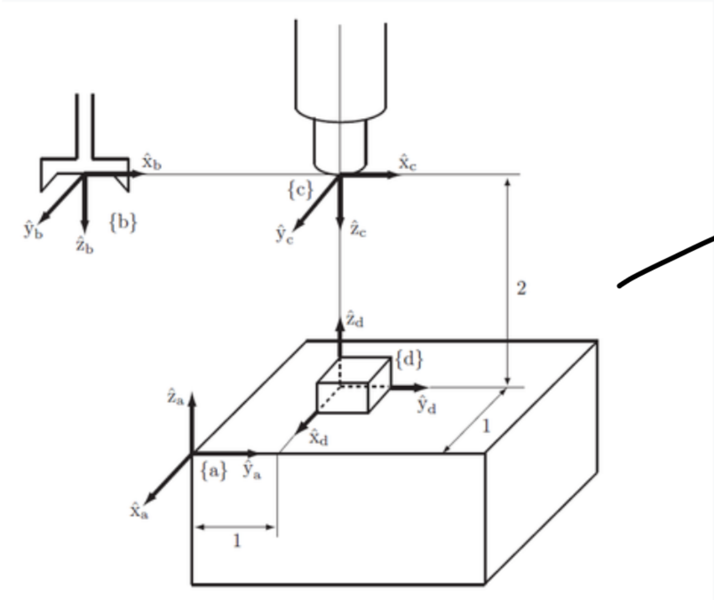


Figure 3.23: Four reference frames defined in a robot's workspace.

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Four reference frames are shown in the robot workspace of Figure 3.23: the fixed frame {a}, the end-effector frame {b}, the camera frame {c}, and the workpiece frame {d}.

(a) Find T_{ad} and T_{cd} in terms of the dimensions given in the figure.
(b) Find T_{ab} given that

$$T_{bc} = \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Frame not rotated

translation vector

a)

$$T_{ad} = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{cd} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

b) $T_{ab} = T_{ad} \cdot T_{dc} \cdot T_{cb}$

$$T_{cb} = T_{bc}^{-1} = \begin{bmatrix} R^T & -R^T\vec{p} \\ \vec{0}^T & 1 \end{bmatrix}$$

$$T_{cB} = \begin{bmatrix} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$R^T \cdot p = [0\ 0\ -2]^T$ but then negate it

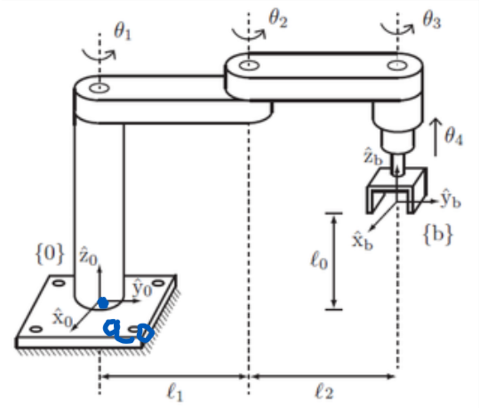
$$T_{dc} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{ad} = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{dc}T_{cB} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & -4 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\boxed{T_{ab} = \begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & -3 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}}$$

Problem 2



The RRRP SCARA robot in the figure is in its zero position. Determine the end-effector zero position configuration M, the screw axes S_i in 0, and the screw axes B_i in b. For $\ell_0 = \ell_1 = \ell_2 = 1$ and the joint variable values $\theta = (0, \pi/2, -pi/2, 1)$, use both the body form and the spatial form of the product of exponentials formula to find the transformation T.

XYZ XYZ

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & L_1 + L_2 \\ 0 & 0 & 1 & L_0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$S_1 : \omega_1 = (0,0,1)^T \quad V_1 = (0,0,0)^T$$

$$S_1 = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{matrix} i & j & k \\ 0 & 0 & 1 \\ 0 & L_1 & 0 \end{matrix}$$

$x L_1$

$-\omega \times q$

$$S_2 : \omega_2 = (0,0,1)^T \quad V_2 = (L_1, 0, 0)$$

$$S_2 = \begin{bmatrix} 0 & -1 & 0 & L_1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathcal{S}_3 : \omega_3 = (0, 0, 1)^T \quad V_3 = (L_1 + L_2, 0, 0)^T$$

$$\mathcal{S}_3 : \begin{bmatrix} 0 & -1 & 0 & 2 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

prismic

$$\mathcal{S}_4 : \omega_4 = (0, 0, 0)^T \quad V_4 = (0, 0, 1)^T$$

$$\mathcal{S}_4 : \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$B_i = Ad_{M^{-1}} \mathcal{S}_i \qquad T^{-1} = \begin{bmatrix} R^T & -R^T p \\ 0 & 1 \end{bmatrix}$$

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad M^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{dT} = \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix}$$

R is identity

$$\tilde{p} = \begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix} \cdot R = \hat{p}$$

$\therefore$

$$A_{dM^{-1}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 & 0 \\ 2 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{matrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{matrix}$$

put in vectorform

$$B_1 = A_{dM^{-1}} \cdot S_1 = (0, 0, 1, -2, 0, 0)^T$$

$$B_2 = (0, 0, 1, -1, 0, 0)^T$$

$$B_3 = (0, 0, 1, 0, 0, 0)^T$$

$$B_4 = (0, 0, 0, 0, 0, 1)^T$$

$$T(\theta) = e^{[S_1]\theta_1} e^{[S_2\theta_2]} e^{[S_3\theta_3]} e^{[S_4\theta_4]} M$$

$$e^{[S]\theta} =$$

$$\begin{bmatrix} e^{[w]\theta} & (I\theta + (1 - c\theta)[w] + (\theta - s\theta)[w]^2)v \\ 0 & 1 \end{bmatrix}$$

$$e^{[w]\theta} = I + s\theta[w] + (1 - c\theta)[w]^2$$

$$\theta_1 = 0$$

$$e^{[S_1]0} = I$$

$$\theta_2 = \pi/2$$

$$\tilde{w} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{2} \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$e^{[w]\pi/2} = I + [w] + [w]^2$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\left(I\,\pi/2 + [w] + \left(\frac{\pi}{2} - 1\right)[w]^2\right) v_2$$

$(\pi/2, 0, 0)^T$

$(0, 1, 0)^T$

$1 - \frac{\pi}{2}, 0, 0$

$$\boxed{\therefore\ e^{[S_2]\frac{\pi}{2}} = \begin{bmatrix} 0 & -1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}$$

$\theta_3 = -\pi/2$ $\quad w = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{2} \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$e^{-[w]\pi/2} = I - [w] + [w]^2$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$2, 0, 0$

T

$$\left(-I\,\pi/2 + [w] + \left(-\frac{\pi}{2} - 1\right)[w]^2\right) v_3$$

$-2, 0, 0$

$$(-\pi, 0, 0)^T + (0, 2, 0)^T + (\pi+1, 0, 0)^T$$

$$= (-2, 2, 0)^T$$

prismatic joint shortcut

$$e^{[S]\theta} = \begin{matrix} I & v\theta \\ 0 & 1 \end{matrix}$$

001

$$\boxed{e^{-[S_3]\pi/2} = \begin{bmatrix} 0 & 1 & 0 & -2 \\ -1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}$$

$$\boxed{e^{[S_4]} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}}$$

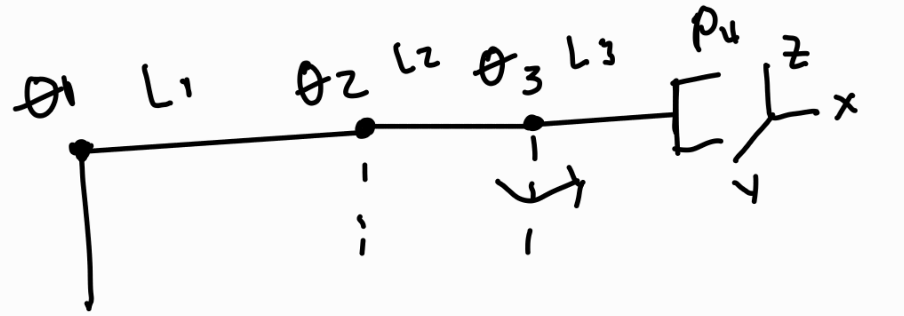
$$T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & -2 \\ -1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{T_s = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} = T_b}$$

Problem 3

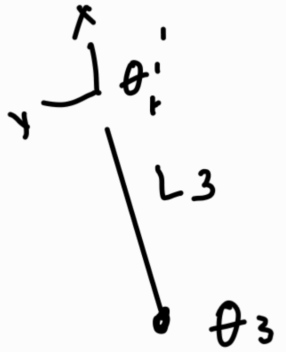
Write a program that solves the analytical inverse kinematics for a planar 3R robot with link lengths $L_1 = 3, L_2 = 2$, and $L_3 = 1$, given the desired position (x, y) and orientation θ of a frame fixed to the tip of the robot. Each joint has no limits. Your program should find all the solutions, give the joint angles for each, and draw the robot in these configurations. Test the code for the case of $(x, y, \theta) = (4, 0.5, 22°)$ and the case $(x, y, \theta) = (1.5, -1.5, -105°)$.

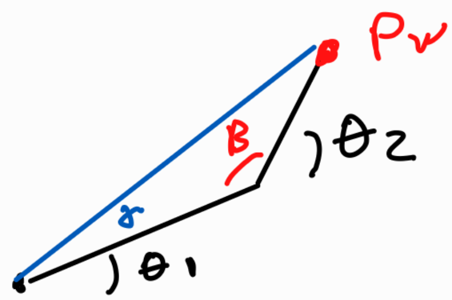
$P_w =$ wrist



x, y, θ

$$\vec{P}_w = \vec{P}_E - L_3 \cos\theta \, \hat{x} + L_3 \sin\theta \, \hat{y}$$





$\therefore$

$$L_1^2 + L_2^2 - 2L_1L_2\cos(\beta) = |P_w|^2$$

$$\beta = \cos^{-1}\left(\frac{L_1^2 + L_2^2 - P_w^2}{2L_1L_2}\right)$$

$$\alpha = \cos^{-1}\left(\frac{P_w^2 + L_1^2 - L_2^2}{2L_1|P_w|^2}\right)$$

$\gamma = \text{arctan2}(P_w)$

$$\theta_1 = \gamma - \alpha \qquad \theta_2 = \pi - \beta$$